Pratik Rauniyar

Certifications and Work samples

Certifications

CERTIFICATE of INTERNSHIP

Computer Application Development and Data Science

THIS CERTIFICATE IS AWARDED TO

Pratik Rauniyar

BY THE UNIVERSITY OF IDAHO
INTERMOUNTAIN FORESTRY COOPERATIVE
for exemplary completion of a competitive National Science Foundation
Research Experience for Undergraduates
ON THIS DAY OF AUGUST 16, 2024

Mark Kimsey
IFC, DIRECTOR

THE UNIVERSITY OF IDAHO 1889

Intermountain Forestry Cooperative

Verified
Certificate

IBM

Rav Ahuja
AI and Data Science Program Director
IBM

This is to certify that

prateek rauniyar

successfully completed and received a passing grade in

DB0201EN: SQL for Data Science

a course of study offered by IBM, an online learning initiative of IBM.

edX

Verified Certificate
Issued June 26, 2024

Valid Certificate ID
742e2f851cef479b87f91cfe877c6fce

UCDAVIS

COURSE CERTIFICATE

Mar 16, 2022

PRATIK RAUNIYAR

has successfully completed

SQL for Data Science

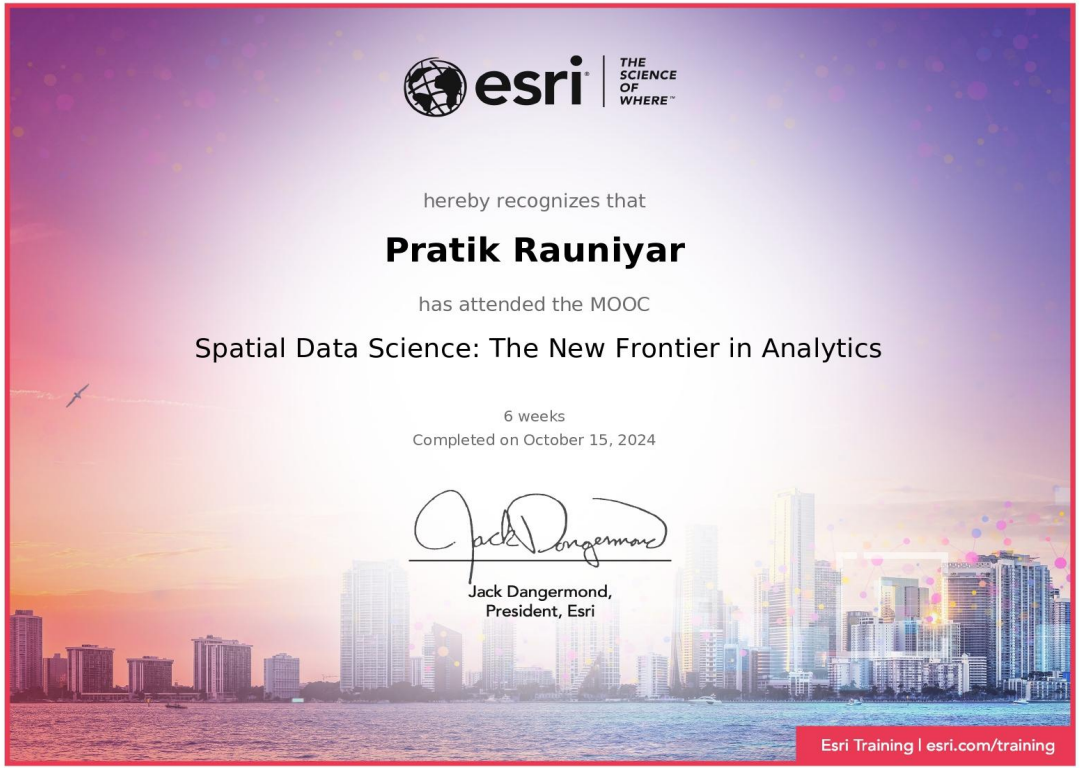
an online non-credit course authorized by University of California, Davis and offered through Coursera

EDUCATION FOR EVERYONE
coursera
COURSE CERTIFICATE

Sadie St. Lawrence
Founder/CEO Human Machine Collaboration Institute & Founder of Women in Data
Instructor, Continuing and Professional Education

Verify at:
https://coursera.org/verify/BMSNX66GC42D

Coursera has confirmed the identity of this individual and their participation in the course.


esri
THE SCIENCE OF WHERE
hereby recognizes that
Pratik Rauniyar
has attended the MOOC
Spatial Data Science: The New Frontier in Analytics
6 weeks
Completed on October 15, 2024
Jack Dangermond,
President, Esri
Esri Training | esri.com/training

DeepLearning.AI

Stanford ONLINE

Aug 15, 2024

PRATIK RAUNIYAR

has successfully completed

Supervised Machine Learning: Regression and Classification

an online non-credit course authorized by DeepLearning.AI and Stanford University and offered through Coursera

Andrew Ng, Instructor, DeepLearning.AI

COURSE CERTIFICATE

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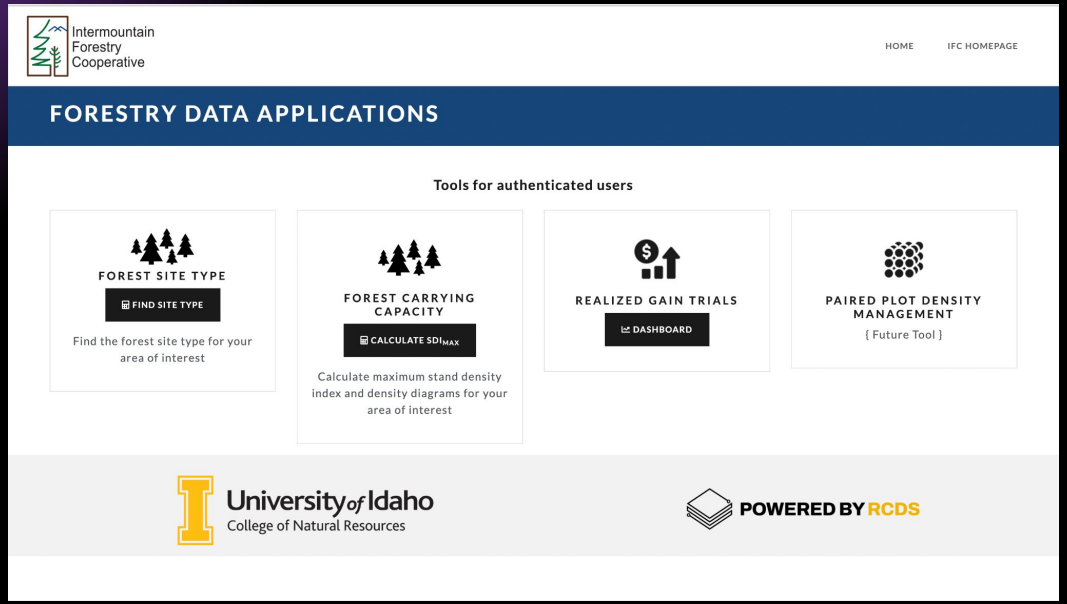
Verify at:
https://coursera.org/verify/VDULR30U389G

Coursera has confirmed the identity of this individual and their participation in the course.

Work samples

Inter-Mountain Forestry Cooperative(IFC), University of Idaho, intern (May, 2024 - Aug 2024)

Demo : https://youtu.be/gj_Fzifr6mo
Github : https://github.com/prateekrauniyar345/ifc-dev



What it Does

- Provides a web tool for forestry researchers to upload/draw polygons and get summary statistics from regional raster layers.
- Calculates Maximum Stand Density Index (SDImax) using user inputs (QMD, PBA) and regional equations.
- Supports both Inland and Westside base layers, adapting equations to user-provided forest composition data.

How it Does

- Frontend: Web form captures polygons and inputs (QMD, PBA) and dynamically generates constants (constVal) to simplify backend processing.
- Backend (Flask): Applies SDImax equations to raster layers (GeoTIFFs) and computes results per polygon.
- Environment: Runs on Flask virtual environment; hosted on ifc-dev.nkn.uidaho.edu with WSGI for deployment.

Tech Stack

- Backend: Flask (Python 3.8), WSGI
- Data: GeoTIFF raster layers, SDImax forestry equations
- Frontend: HTML templates (Jinja2), web forms
- Database/Config: Configurable environment (config.py, requirements.txt)
- Deployment: Virtualenv, Linux server (UIDAHO NKN infrastructure), Gunicorn/WSGI processes

Adobe Express Hackathon - link

Demo : https://youtu.be/PgglycNCtx8
Github : https://github.com/prateekrauniyar345/adobe-first-add-on
Try : https://adobesparkpost.app.link/TR9Mb7TXFLb?mode=private&claimCode=wk63mikg8:RZK9O05J

What it Does

- Lets users **upload CSV/XLSX files** and receive **AI-powered chart recommendations**.
- Generates **interactive charts** with real-time style customization.
- Seamlessly **inserts visualizations into Adobe Express documents** or exports as PNG.

How it Does

- **Frontend (React 18+):** Handles file upload, chart previews, and style customization.
- **Charting (Observable Plot):** Creates bar, line, scatter, histogram, box, and heatmap charts with dynamic styling.
- **Adobe Integration:** Uses Adobe Express Document Sandbox Runtime API to insert charts directly into documents.

Tech Stack

- **Frontend:** React 18+, Bootstrap CSS, Webpack, Babel
- **Charting & Data:** Observable Plot, Papa Parse (CSV), SheetJS (Excel)
- **Adobe Integration:** Adobe Express Document Sandbox Runtime API, manifest.json config
- **AI Services:** LLM-based chart recommendation service (serverless function)
- **Development Tools:** Node.js, npm, React hooks (useState, useEffect, useMemo, useCallback)

Work Sync AI - link

Demo : https://youtu.be/MPo8DQRLKSI
Github : https://github.com/prateekrauniyar345/workspace_creator

What it Does

- Lets users **create and manage smart workspaces** with apps, URLs, and custom window layouts.
- Features an **AI-powered profile generator** that builds workspaces from natural language prompts.
- Automates **app launching, website opening, and workspace scheduling** for improved productivity.

How it Does

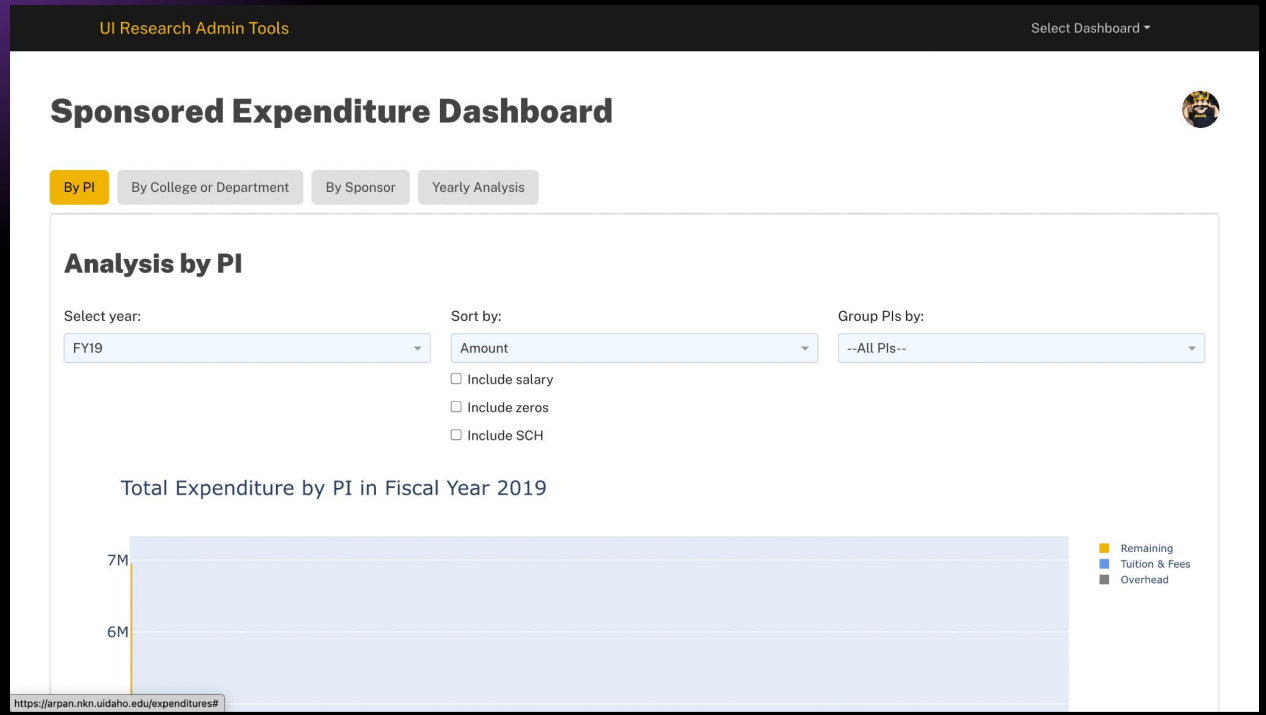
- **AI (Nebius LLM):** Parses user prompts into structured profiles of apps + URLs.
- **Automation (Screenpipe):** Opens applications, launches websites, and arranges windows automatically.
- **Frontend (Next.js + React):** Provides UI for profile management, layout editing, and scheduling.

Tech Stack

- **Frontend/UI:** Next.js App Router, React, TailwindCSS
- **AI:** Nebius AI Studio (LLaMA 3.1 70B-Instruct, OpenAI-compatible API)
- **Automation:** Screenpipe CLI / API for app & window management
- **Runtime/Tools:** Bun (package manager), macOS Spotlight (installed app fetch)

Expenditure Dashboard - IIDS/ RCDS (Aug 2024 - Current)

Demo : https://youtu.be/B30J1-uMZ4w
Github : https://github.com/ui-insight/dashboard



What it does:

1. **Research Data Visualization & Analytics** - Provides interactive dashboards for analyzing sponsored expenditures, grant workloads, and research funding data across departments, colleges, PIs, and sponsors

2. **Pre-award Proposal Management** - Automates proposal due date calculations, generates tiered support emails, manages SPA (Sponsored Programs Administrator) assignments, and tracks proposal deadlines with holiday considerations

3. **User & Access Management** - Implements role-based access control for different research administration stakeholders with user profiles, group management, and secure authentication through Azure AD

How it does it:

1. **Interactive Dashboard Framework** - Uses Dash/Plotly to create dynamic, filterable visualizations with real-time data updates, allowing users to drill down into expenditure patterns, PI performance, and departmental analytics

2. **Database-Driven Workflow Automation** - Connects to MySQL databases to fetch research data and automatically generates templated emails for proposal management workflows, calculating business day deadlines and managing holiday schedules

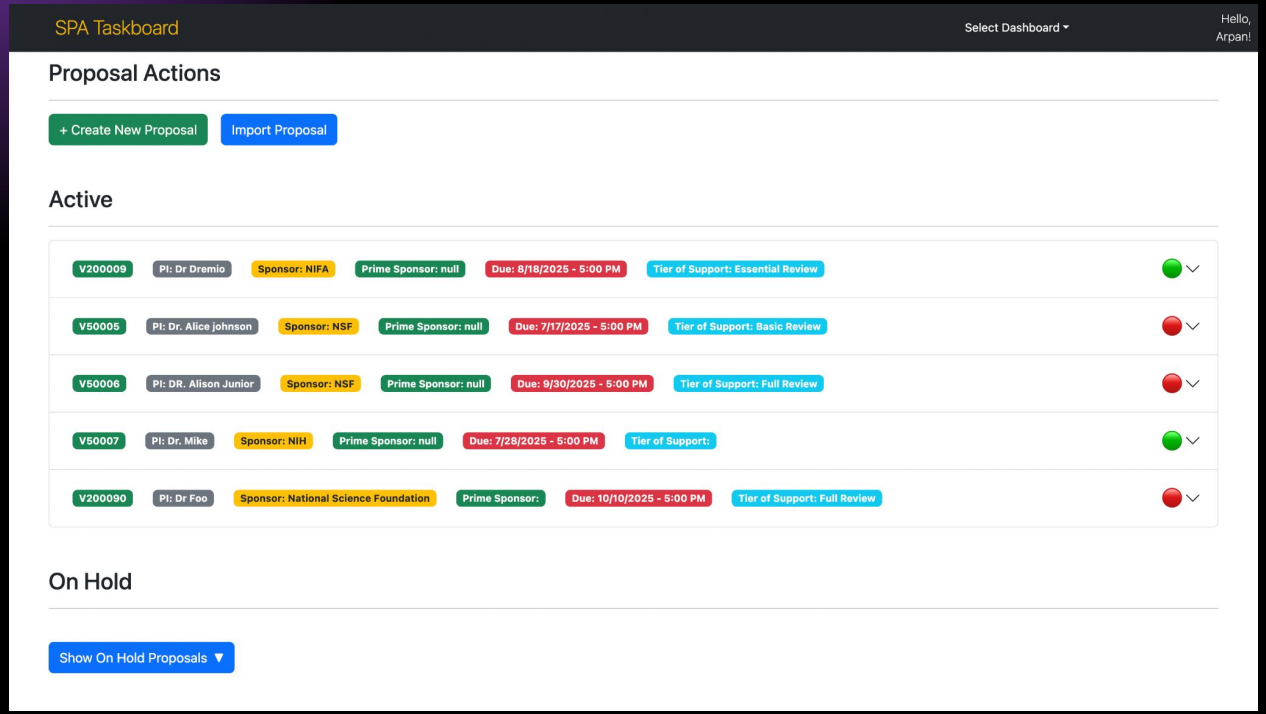
3. **Multi-layered Authentication & Authorization** - Integrates Flask-Login with Azure OAuth for secure user authentication, implements granular role-based permissions, and maintains user sessions with profile management capabilities

Tech Stack:

- **Backend**: Python, Flask, SQLAlchemy, MySQL
- **Frontend**: Dash, Plotly, HTML/CSS, Bootstrap, JavaScript
- **Authentication**: Azure AD OAuth, Flask-Login
- **Database**: MySQL (production), SQLite (local)
- **Deployment**: Gunicorn, Heroku (based on Procfile)

SPA Taskboard - IIDS/ RCDS (Aug 2024 - Current)

Demo : https://youtu.be/XexdCSOUQak
Github : https://github.com/ui-insight/reactfast



What it Does

- SPA Taskboard, a research admin tool made for the Office of Sponsored Program for the University of Idaho, used to create, track and adjust the proposal submitted by PI/faculty.
- Allows the SPA to track, create, import and export the proposals with their relevant documents as well as ability to create the proposal directly from Due Date Calculator.
- React frontend and FastAPI backend that manages the entire research proposal lifecycle from creation to submission, with MySQL/MariaDB database storage.

How it Does

- Backend uses FastAPI with modular folders for APIs, models, services, utilities, and SQL schema/triggers for database integrity.
- Frontend built with React using reusable components, pages, and API wrappers for backend communication.
- Git workflow enforces branch protection, pull requests, CI tests, and reviews before merging into production.

Tech Stack

- Backend: FastAPI, Python, Pydantic, Pytest, MySQL/MariaDB
- Frontend: React.js, JavaScript/TypeScript (optional)
- DevOps/Workflow: GitHub (protected branches, PR reviews, CI/CD), environment-based configs (.env.dev, .env.test, .env.prod)

Due Date Calculator - IIDS/ RCDS (Aug 2024 - Current)

Demo : https://youtu.be/JVppuv5-Rpo
Github : https://github.com/ui-insight/reactfast

Due Date Calculator

Select Dashboard

Hello, Arpan!

Due Date & Time:

09/27/2025 05:00 PM

Tier of Support

Full Review

Proposal created Date & Time:

09/27/2025 05:00 PM

Assign SPA

Arpan Pal (apal@uidaho.edu)

Assign SPA

Relevant Due Dates

Business Days Prior	Date	Time
10 business days prior	9/15/2025	08:00 AM
6 business days prior	9/19/2025	08:00 AM
6 business days prior	9/19/2025	08:00 AM

Initial Email

This proposal was entered into VERAS with less than four-business days to the sponsor deadline, and/or the documents are not in final form for OSP review and approvals. A Tiered Proposal Support - Deadline Waiver Request will be required for the OSP to proceed with an essential review, and only after the SPA has handled all other proposals with similar due dates

What it does:

1. Calculates proposal review deadlines based on different service tiers and business hours
2. Manages holidays to ensure accurate business day calculations
3. Generates automated email templates for administrators with calculated dates

How it does:

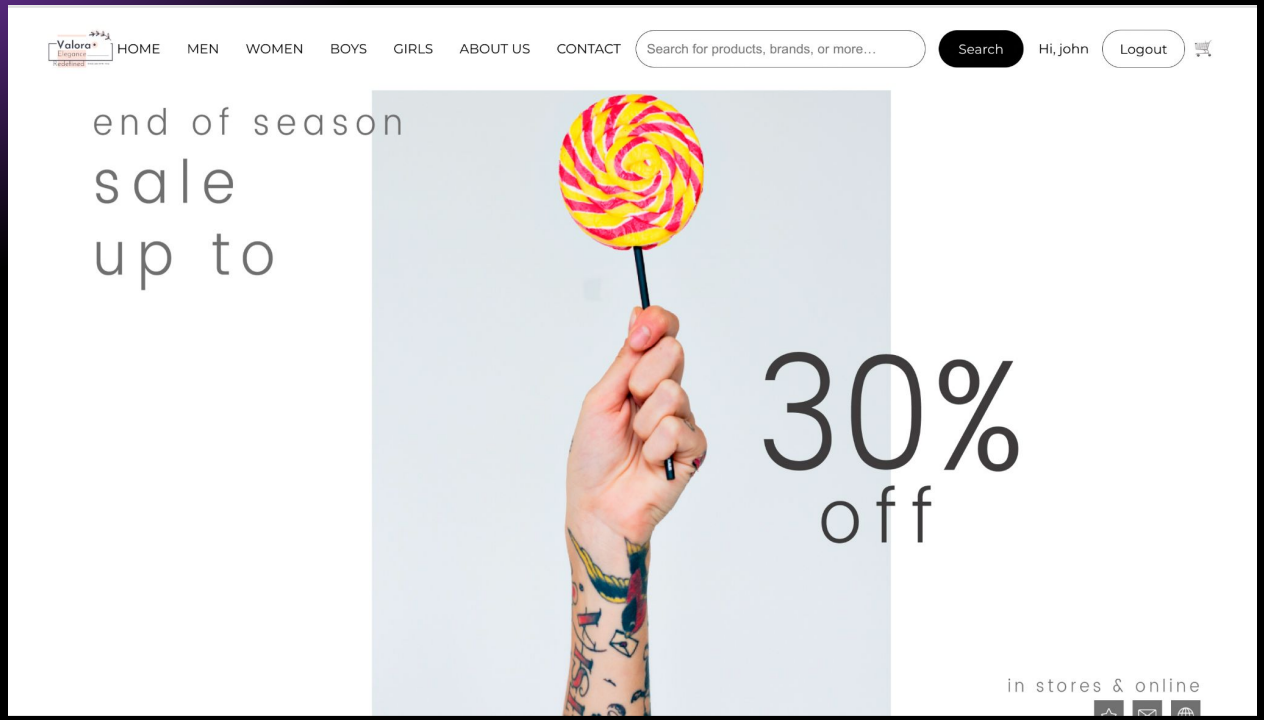
1. Business hours engine - Subtracts working hours from due dates excluding weekends/holidays
2. Smart templating - Auto-populates emails based on timing and review requirements
3. Interactive dashboard - Real-time UI with date pickers, dropdowns, and rich text editing

Tech Stack:

- Frontend: React + Vite
- UI: Mantine components, TipTap editor, Bootstrap
- Utils: date-fns, date-holidays library
- Tools: ESLint, CSS modules

Valora

Demo : https://youtu.be/_KEk4hCkygA
Github : https://github.com/prateekrauniyar345/Valora
Try : https://valora-xi.vercel.app/



What it Does

- Provides a **full-stack e-commerce platform** with user authentication, product catalog, cart, and checkout.
- Supports **secure payments via Stripe** and role-based access (customer vs. admin).
- Delivers a **modern responsive UI/UX** with real-time search, order history, and admin product management.

How it Does

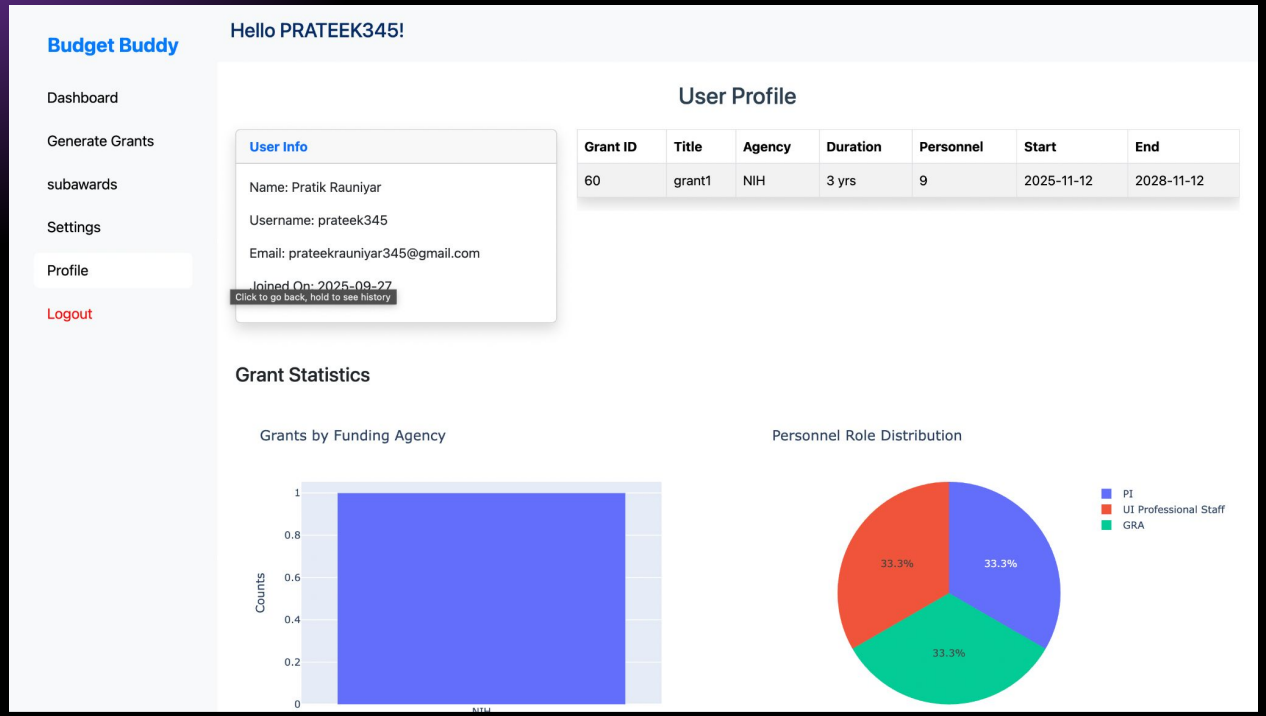
- **Frontend (React + Vite):** Uses React Router, reusable components, and responsive design with CSS modules.
- **Backend (Node.js + Express):** Implements RESTful APIs, session-based authentication, and secure order/payment handling.
- **Database (MongoDB + Mongoose):** Manages users, products, carts, and orders with flexible NoSQL schemas.

Tech Stack

- **Frontend:** React 19, Vite, React Router DOM, React Icons, React Slick, ESLint
- **Backend:** Node.js, Express.js, MongoDB, Mongoose, Express Session, CORS, Stripe, Dotenv
- **Development Tools:** Nodemon, ESLint, Vite Dev Server
- **Security:** Environment variables, secure cookies, role-based access, Stripe integration.

Grant Budget Generator

Demo : https://youtu.be/XBuNHSGdXzk
Github : https://github.com/prateekrauniyar345/grant-budget-manager
Try : https://grant-budget-manager.onrender.com/



What it Does

- Helps researchers build compliant multi-year NSF/NIH budgets with personnel, fringe, travel, and expense tracking.
- Provides interactive dashboards for grant planning, subawards, and analytics.
- Allows secure export of budgets to Excel for proposal submission.

How it Does

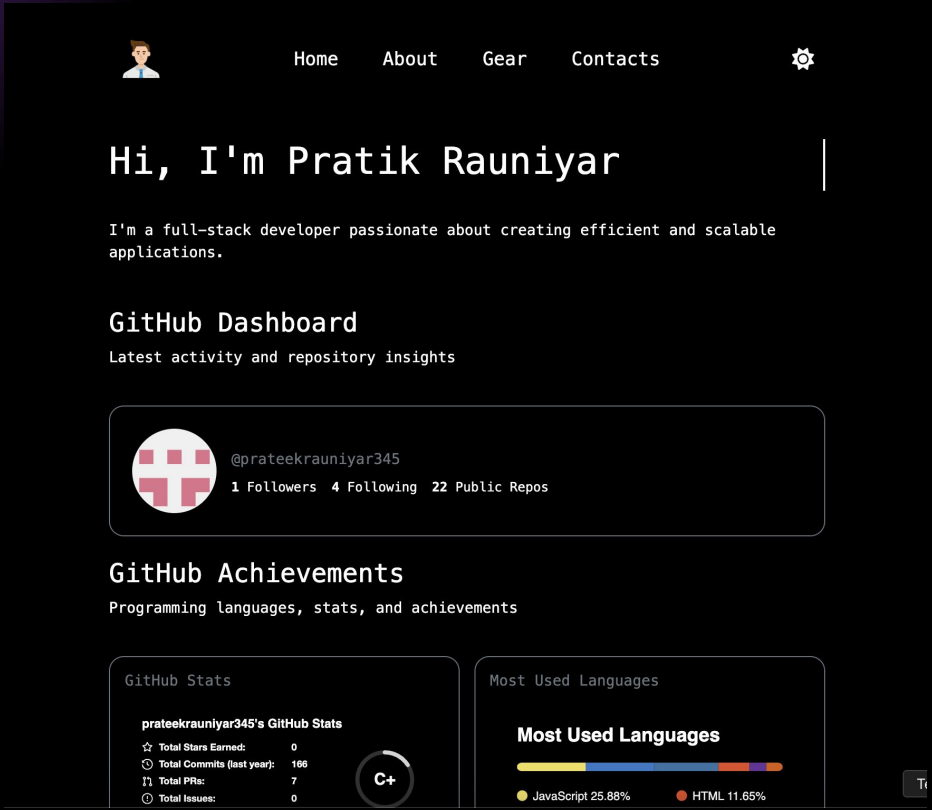
- Uses Flask for authentication, user flows, and serving the Dash app.
- Leverages Dash (Plotly) for interactive pages like dashboard, grant generator, and subawards.
- Stores and manages data with PostgreSQL + SQLAlchemy, with support for MySQL → Postgres migration.

Tech Stack

- Backend: Flask, SQLAlchemy, PostgreSQL, Gunicorn
- Frontend/UI: Dash (Plotly), Jinja2 templates, CSS/JS assets
- Auth & Security: Flask-Login, Bcrypt, environment-based secrets
- Deployment: Render, Gunicorn, environment variables
- Tools: ReportLab (PDF/Excel export), data migration scripts, virtual environments

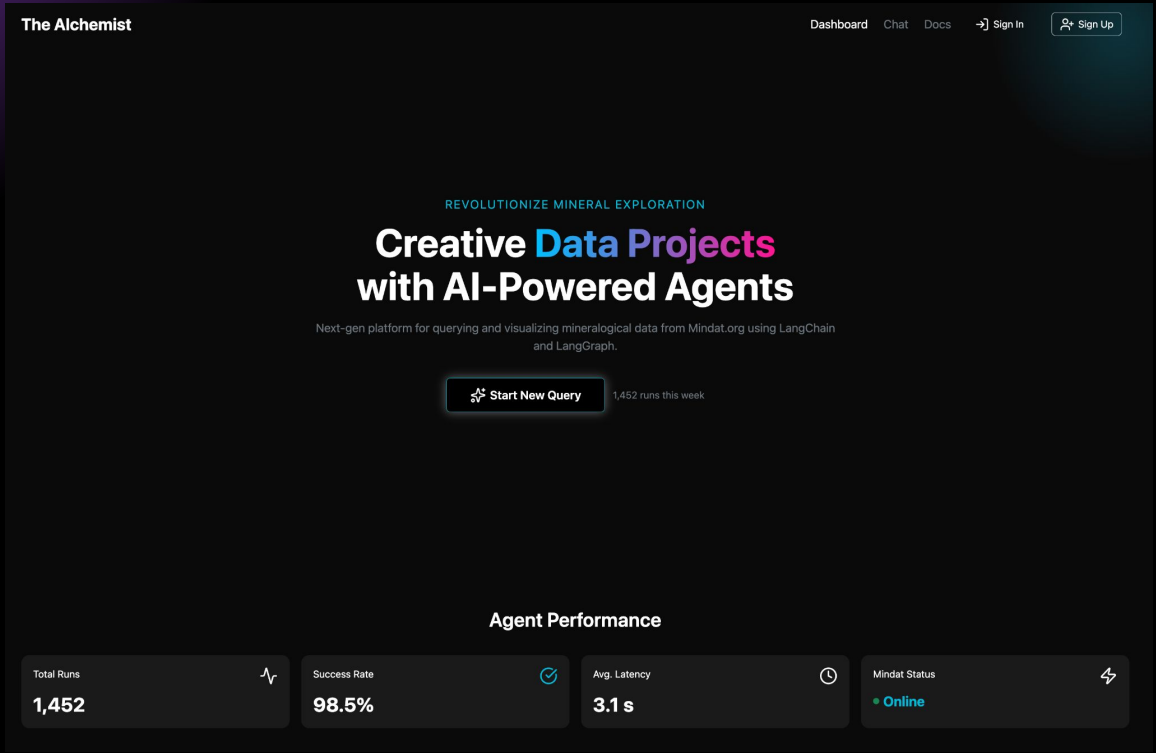
Portfolio

Github : https://github.com/prateekrauniyar345/new-portfolio.git
Try : https://portfolionew-fawn.vercel.app/



The Alchemist

Github :
https://github.com/prateekrauniyar345/Senior-Design-The-Alchemist



What it Does

- **Provides** an intelligent, AI-powered system for exploring and analyzing mineral data from Mindat.org.
- **Supports** complex natural language queries to filter minerals by properties such as hardness, crystal systems, and chemical elements.
- **Delivers** advanced data visualizations including element distribution histograms, mineral relationship network diagrams, and geographical heatmaps for mineral localities.

How it Does

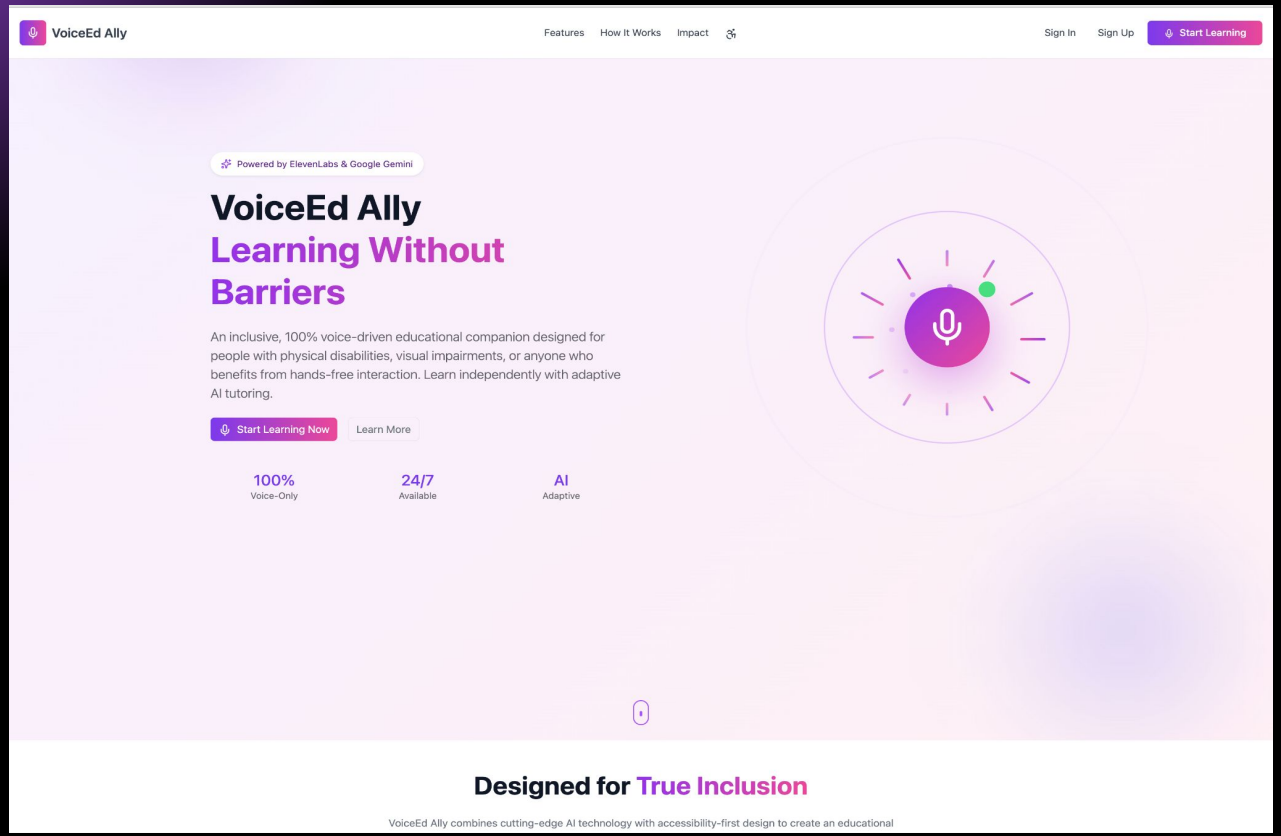
- **Multi-Agent Workflow (LangChain + LangGraph):** Utilizes a supervisor agent to route user queries to specialized agents (Collector, Plotter, etc.) for data retrieval and processing.
- **Backend (FastAPI):** Implements RESTful endpoints to handle asynchronous communication between the AI agents and the frontend.
- **Monitoring & Debugging (LangSmith):** Integrates full tracing of agent behavior to track execution flow and optimize multi-step AI reasoning.

Tech Stack

- **AI & Logic:** LangChain, LangGraph, LangSmith, Tavily Search
- **Backend:** FastAPI, Uvicorn, Python 3.10+
- **Data & Visualization:** Matplotlib, Plotly, Mindat API
- **Frontend:** FastAPI Templates (HTML/JS/CSS)
- **Development Tools:** Pip, Dotenv, Python REPL Tool

VoiceED Ally

Github : https://github.com/prateekrauniyar345/AI-Partner-Catalyst-Accelerate-Innovation



What it Does

- **Inclusive Educational Companion:** Provides a fully voice-driven Web App (WA) designed specifically for users with physical, motor, or visual impairments to learn independently.
- **Adaptive Learning:** Utilizes AI to detect user sentiment and fatigue from voice cues, automatically adjusting the teaching pace, tone, and content to provide a supportive environment.
- **Multimodal Accessibility:** Delivers academic and life-skills tutoring through natural speech, complemented by visual enrichment (images and educational Shorts) with spoken descriptions.
- **Community Knowledge Vault:** Features an optional repository where users can share and access anonymized tips for independent living and accessibility advocacy.

How it Does

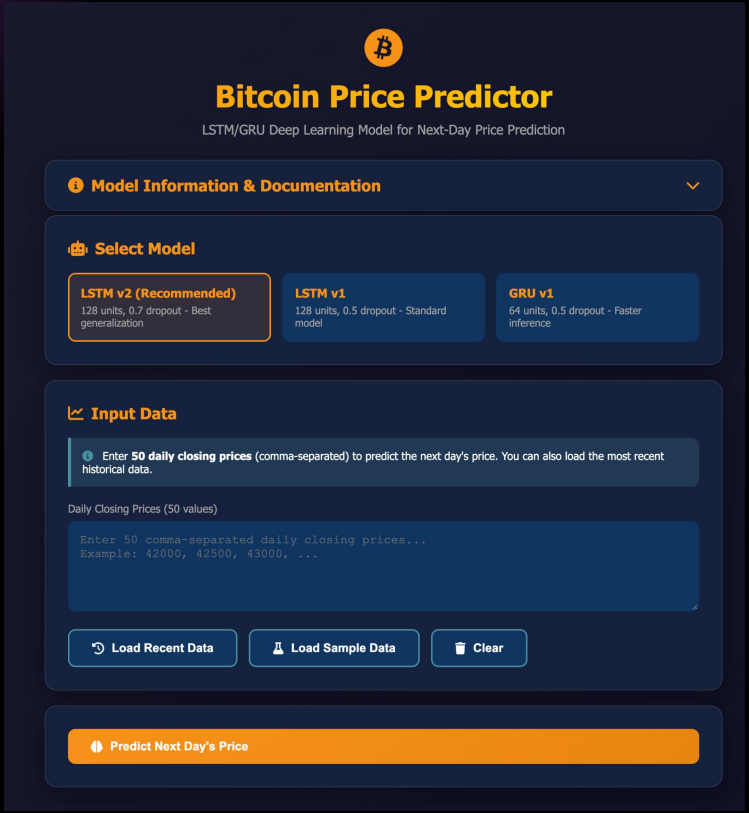
- **Real-Time Voice Intelligence (ElevenLabs):** Employs ElevenLabs Conversational AI Agents for high-quality, empathetic text-to-speech and dynamic voice personalities.
- **Sentiment & Reasoning (Vertex AI Gemini):** Leverages Google Cloud's Gemini models to perform sentiment analysis on user input and generate personalized, context-aware educational explanations.
- **Content Integration (Pexels & YouTube APIs):** Dynamically retrieves royalty-free images and educational video content to provide a rich, interactive visual layer to the voice-first experience.
- **PWA Architecture:** Built as a Progressive Web App to ensure the platform is installable on mobile devices and remains accessible for hands-free use anywhere.

Tech Stack

- **AI & Logic:** Google Vertex AI Gemini (Flash/Pro), ElevenLabs Conversational AI & SDK
- **Frontend:** React, Vite, TypeScript, Tailwind CSS, Vite PWA Plugin
- **Backend:** Python, Flask REST API
- **Data & Media:** Firebase (Firestore/Storage), YouTube Data API v3, Pexels API
- **Development Tools:** Node.js, Git, Pip, Dotenv

Bitcoin Price Predictor

Github : https://github.com/prateekrauniyar345/Bitcoin-Price-Prediction-Model



What it Does

- **Time-Series Forecasting:** Predicts the next day's Bitcoin price using historical daily closing data.
- **Multiple Model Selection:** Offers three distinct deep learning architectures (LSTM v1, LSTM v2, and GRU v1) to compare prediction accuracy and speed.
- **Interactive Interface:** Provides a modern, responsive web dashboard for users to input price data and visualize results.
- **Programmatic Access:** Features a REST API for integration with external applications and a CLI for batch processing.

How it Does

- **Deep Learning Architectures:** Utilizes **LSTM (Long Short-Term Memory)** to capture long-term patterns and **GRU (Gated Recurrent Unit)** for faster inference with fewer parameters.
- **Data Pipeline:** Implements a sliding window approach with a **50-day lookback** period; it normalizes raw data using **MinMaxScaler** to improve training efficiency.
- **Model Optimization:** Employs the Adam optimizer, Mean Squared Error (MSE) loss function, and techniques like **Batch Normalization** and high **Dropout rates (up to 70%)** to prevent overfitting.
- **Web & API Serving:** A Flask-based backend handles model inference and serves both the HTML frontend and JSON-based API endpoints.

Tech Stack

- **AI & Logic:** TensorFlow/Keras, NumPy, Scikit-learn (MinMaxScaler)
- **Backend:** Python 3.10+, Flask (REST API), Werkzeug
- **Data & Processing:** Pandas, Joblib (for scaler persistence)
- **Frontend:** HTML, CSS, JavaScript (Visualizations)
- **Development Tools:** Jupyter Notebooks, Git, Pip, H5PY (model storage)